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De poolcoördinaten van $(4, -4)$; (r, θ) —?

$$r = \sqrt{x^2 + y^2} \Rightarrow r = \sqrt{4^2 + (-4)^2} = \sqrt{32} = 4\sqrt{2}$$

$$\tan \theta = \frac{-y}{x} \Rightarrow \tan \theta = \frac{-4}{4} = -1 \Rightarrow \theta = \tan^{-1}(-1) = -\frac{\pi}{4} \Rightarrow \theta = 2\pi - \frac{\pi}{4}$$

$$\Rightarrow \theta = \frac{8\pi}{4} - \frac{\pi}{4} = \frac{7\pi}{4}$$

Resultaat: $(r, \theta) = \left(4\sqrt{2}, \frac{7\pi}{4}\right)$

References